

Jointly Optimised End-to-End PPO for Regime-Aware Portfolio Management

Colombo Stock Exchange (CSE) -- Banking Sector

Project Description and Methodology

Date: 03 August 2026

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Problem Statement

Traditional quantitative portfolio management typically treats "market regime detection" (identifying bull/bear/sideways conditions) and "portfolio allocation" as two separate stages: first cluster historical data into regimes using an unsupervised method (e.g. Hidden Markov Models, k-means), freeze that model, then train a separate allocation strategy conditioned on those fixed regime labels.

This project's central methodological claim is that this two-stage separation is unnecessary and potentially suboptimal -- regimes discovered for their own sake (to minimise a clustering objective) are not guaranteed to be the regimes that actually help a portfolio perform better.

Objective

Build a single, end-to-end differentiable system where regime detection and portfolio allocation are learned jointly, by the same training signal (portfolio risk-adjusted return), in the same backward pass -- so the model discovers whatever partition of market conditions actually improves capital allocation, rather than a partition imposed by an unrelated objective.

Data

- Source: Colombo Stock Exchange (CSE), banking sector, daily OHLCV data

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Methodology

1. Feature Engineering and Windowing

Each of the 10 assets is expanded into 15 stationary technical indicators per day; every trading decision is based on a rolling 30-day window of history (not a single day), since a "regime" is a property of a stretch of time, not an isolated observation.

2. Differentiable Regime Encoder (the core contribution)

A shared neural pipeline, applied identically to every asset:

- Variable Selection Network (VSN): learns a soft, time-varying importance weighting over the 15 indicators at every timestep (interpretable -- e.g. leaning on volatility features during a crash, momentum features in a trend)

3. Actor-Critic Policy

A shared per-asset actor network converts each asset's latent state into a Gaussian-sampled, softmax-normalised portfolio weight; a critic estimates the value of the current market state to stabilise training.

4. Market Environment

Realistic trading constraints: 30% maximum position size per asset, 0.10% transaction cost, 0.05% slippage, daily rebalancing, long-only (no short-selling, since CSE short-sale availability for this account type is unverified).

5. Reward Design

Risk-adjusted, not raw return: a rolling 20-day Sharpe ratio, minus an Entropic Value-at-Risk (EVaR) tail-risk penalty, minus a turnover / transaction-cost penalty.

6. Training

Proximal Policy Optimisation (PPO) with Generalised Advantage Estimation (GAE). Critically, one single backward pass and optimiser step updates the encoder, actor, and critic simultaneously -- this joint update (rather than pre-training the encoder separately) is what makes the regime discovery genuinely "end-to-end".

Evaluation

- Held-out test set backtest, deterministic policy (no exploration noise)

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Honest Limitations

Stated proactively, as they reflect the current state of the work rather than a finished result:

- Single-seed results so far; reinforcement learning outcomes are seed-sensitive, so multi-seed runs

are the natural next step

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